Oleksandr Verhun *Senior Software Engineer*

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SUMMARY

Senior backend engineer with 10+ years of experience, mainly in **Ruby on Rails and Python**, and growing expertise in **Go**. I design and build APIs, event-driven data pipelines and distributed services for high-load systems, and I'm at my best solving performance problems and untangling complex business logic. I make architecture decisions for the systems I own, from data modeling and service boundaries to scalability trade-offs, and see them through to production.

SKILLS

Languages and frameworks: Ruby, Ruby on Rails, Python, FastAPI, Go (Echo), JavaScript (Svelte, Ember, Angular)
Databases: MongoDB (sharded clusters, indexing, aggregation pipelines), PostgreSQL, PostGIS, Redis, SQLite
Messaging and streaming: Kafka, RabbitMQ, Redis (message broker, caching), AWS SQS and SNS
Cloud and infrastructure: AWS (EC2, S3, Lambda, RDS, Elastic Beanstalk, CloudWatch), Docker, Kubernetes, CI/CD pipelines, GitHub Actions
Observability: Prometheus, OpenTelemetry, Grafana, AWS CloudWatch, Rollbar
Testing: RSpec, pytest, Go testing (testify)
Practices: REST and SOAP API design, system design, microservices, event-driven architecture, performance tuning, design docs, code review, knowledge sharing, cross-functional collaboration, on-call and incident response, legacy refactoring
AI: AI-assisted development and code review (coding agents, custom skills), LLM integrations, MCP

EXPERIENCE

Senior Software Engineer, *Corva*
09/2021 – Present | Fully remote
High-load gas/oil analytics platform.
Responsibilities and key contributions:

- Eliminated API and processing-service timeouts at peak load, cutting timeout errors in Rollbar from 5K–30K a day to near zero. Added shard-key filters and targeted indexes, rewrote aggregation pipelines, removed unnecessary sorts and redundant queries, and acquired database connections only when needed.
- Designed and built the subscription system used by all customer companies, with access rules at different levels.
- Built data apps for the real-time drilling data pipeline, giving clients self-service analytics.
- Sped up processing for time-based data apps by redesigning their scheduling, replacing repeated AWS Lambda calls that ran into service limits with a single start/stop call.
- Ran production data fixes that reduced stored data volume.
- Drove observability adoption across services with Prometheus, OpenTelemetry and Grafana, improving uptime and incident response times.
- Own features end to end on a three-person API team, from architecture decisions and design docs through deployment, on-call and incident response.
- Partner with product managers and internal client teams to turn business requirements into technical scope, balancing delivery speed, performance and maintainability.
- Share knowledge across the team through code reviews and design discussions, helping keep the API consistent and maintainable as it grows.

Stack: Ruby on Rails, Python, FastAPI, MongoDB, PostgreSQL, Kafka, Redis, AWS, Docker, Kubernetes, Prometheus, Grafana, OpenTelemetry, RSpec, pytest

Software Engineer, *Brightgrove*
06/2019 – 09/2021
Audit confirmation platform

- Sole API developer on a platform for auditors' year-end confirmation process; owned backend architecture, technical decisions and code quality.
- Built the confirmation workflow API and an auditor-specific document storage system.
- Delivered new features, performance optimizations and production fixes across the backend and front end.

Stack: Ruby on Rails, Svelte

Full Stack Developer, *SQA Solution*
10/2017 – 06/2019
Outstaffed to VerticalResponse, an email marketing platform

- Developed features and fixed production issues in VerticalResponse's email marketing platform, where customers design emails and send campaigns to their contacts.
- Improved application performance, upgraded framework and tooling versions, and wrote automation scripts for team workflows.

Stack: Ruby on Rails, Ember

LANGUAGES

English: B2–C1

Ukrainian: native

Russian: fluent

Polish: basic

Freelance Full Stack Developer, *Upwork*

05/2014 – 10/2017

- Delivered client web and mobile projects end to end, covering front end, back end and deployment.
- Built a real-estate platform (Ruby on Rails, Angular, Ionic) with an email template builder and campaign sender.
- Built a pet listings platform with breeder profiles and e-commerce features.

Stack: Ruby on Rails, Angular, Ionic, RabbitMQ

EDUCATION

Igor Sikorsky Kyiv Polytechnic Institute, *Master's degree, Computer Engineering*

2016 – 2017

Igor Sikorsky Kyiv Polytechnic Institute, *Bachelor's degree, Computer Engineering*

2011 – 2015

PROJECTS

Terminal Fit Recorder, *Go*

Personal project · github.com/vyivrain/terminal_fit_recorder · Bubble Tea, SQLite, GitHub Actions

- Full-screen terminal app for logging workouts and generating AI training plans from each person's history, with isolated per-person profiles and SQLite storage using versioned schema migrations.
- Generates and refines workouts through Claude, Codex or OpenCode with switchable models, also from Telegram via the Hermes AI agent. Imports free-form notes (including Ukrainian) into structured workouts.
- MCP server integration that lets AI assistants read workout history and generate new workouts as tools.
- ~12K lines of Go, a third of it tests. CI runs the suite with the race detector and coverage on every push; tagged releases build checksummed binaries for Linux, macOS and Windows.

Consilium, *Go*

Personal project · github.com/vyivrain/consilium · TCP, Ollama

- Distributed system in which several LLMs deliberate on a question until they reach consensus, built as separate hub, worker and client services.
- The hub coordinates concurrent workers over a custom JSON-over-TCP protocol, with configurable cycle timeouts, agreement thresholds and an optional verification round.
- Each worker runs a different local model through Ollama; consensus is scored by confidence-weighted agreement between models.

Kafka Broker, *Go*

Personal project · github.com/vyivrain/codecrafters-kafka-go · TCP, Kafka wire protocol

- Toy Kafka broker built from scratch through a CodeCrafters challenge. It speaks Kafka's binary wire protocol over TCP and serves concurrent client connections.
- Implements the ApiVersions and DescribeTopicPartitions APIs through a pluggable handler registry, answering topic queries by parsing the KRaft cluster metadata log.

Also in Go: an HTTP/1.1 server and a basic shell, built through CodeCrafters challenges.